

Local convergence rates computed from adjacent refinement levels. The final row shows fitted rates with uncertainties from the log-log least-squares fit. Reynolds number 100.

cycle	# cells	# dofs	L2_velocity	rate	L2_pressure	rate	H1_velocity	rate
0	289	2774	1.110e-04	–	9.059e-04	–	7.770e-03	–
1	1156	10747	7.936e-06	3.81	2.253e-04	2.01	1.328e-03	2.55
2	4624	42299	6.475e-07	3.62	5.626e-05	2.00	2.734e-04	2.28
3	18496	167827	6.542e-08	3.31	1.406e-05	2.00	6.401e-05	2.09
4	73984	668579	7.612e-09	3.10	3.516e-06	2.00	1.572e-05	2.03
fitted rate			–	3.46 ± 0.08	–	2.0021 ± 0.0009	–	2.23 ± 0.06

Table 1: Q2/Q1 elements

cycle	# cells	# dofs	L2_velocity	rate	L2_pressure	rate	H1_velocity	rate
0	289	6633	6.972e-07	–	2.478e-05	–	9.395e-05	–
1	1156	25979	2.965e-08	4.56	3.157e-06	2.97	9.350e-06	3.33
2	4624	102819	1.507e-09	4.30	3.979e-07	2.99	1.068e-06	3.13
3	18496	409091	8.768e-11	4.10	4.993e-08	2.99	1.300e-07	3.04
4	73984	1632003	5.374e-12	4.03	6.254e-09	3.00	1.614e-08	3.01
fitted rate			–	4.24 ± 0.06	–	2.9886 ± 0.0028	–	3.12 ± 0.04

Table 2: Q3/Q2 elements

cycle	# cells	# dofs	L2_velocity	rate	L2_pressure	rate	H1_velocity	rate
0	289	12226	4.634e-09	–	1.640e-07	–	9.531e-07	–
1	1156	48147	1.037e-10	5.48	1.023e-08	4.00	4.949e-08	4.27
2	4624	191083	2.804e-12	5.21	6.416e-10	4.00	2.907e-09	4.09
3	18496	761331	8.987e-14	4.96	4.001e-11	4.00	1.786e-10	4.03
4	73984	3039331	6.668e-14	0.43	2.500e-12	4.00	1.113e-11	4.00
fitted rate			–	4.2 ± 0.6	–	4.0003 ± 0.0006	–	4.09 ± 0.03

Table 3: Q4/Q3 elements